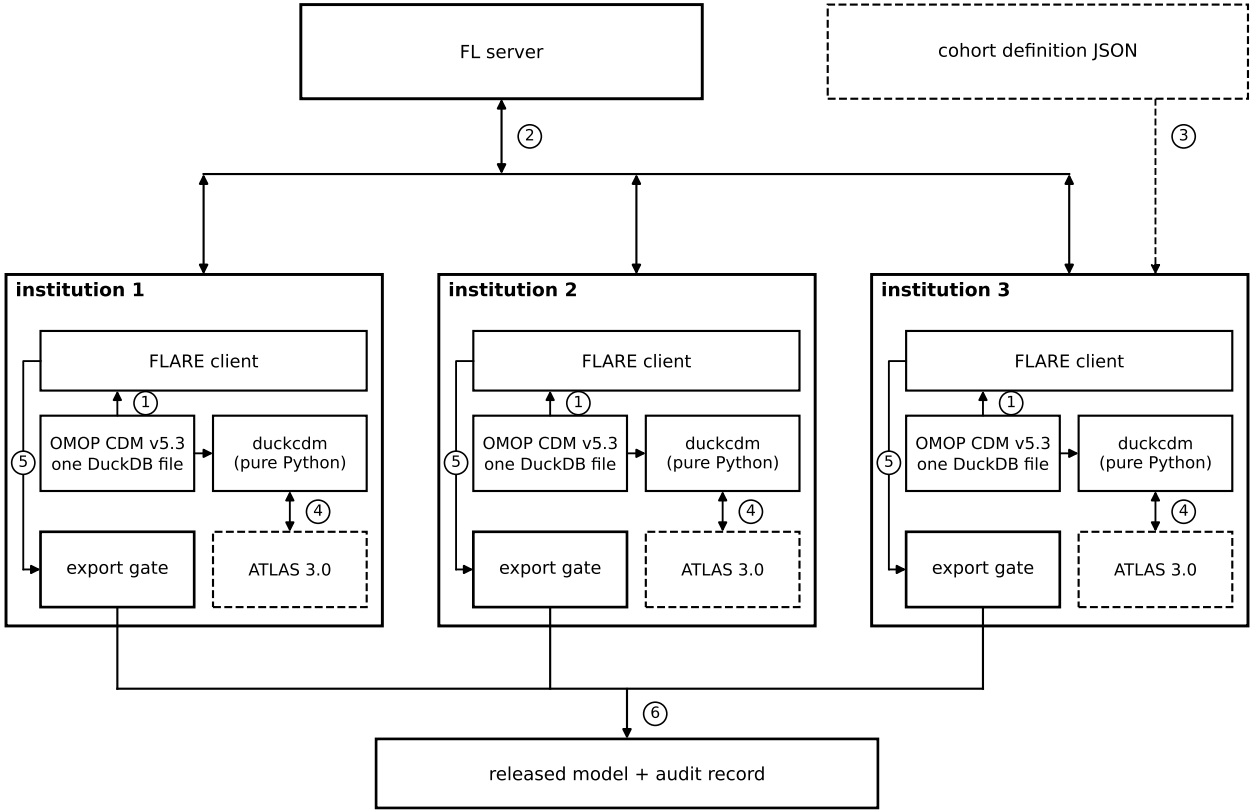

1. cohort and feature matrix read from the file by duckcdm (circe-compiled cohort, SQL features)
2. gradient histograms up, global trees down (NVIDIA FLARE 2.9, histogram-based XGBoost 3.2); no patient rows leave
3. the same cohort definition JSON (institutional ATLAS, circe 1.11.1) is compiled at every node: code goes to the data
4. ATLAS 3.0 in a browser on the same host, served by duckcdm (WebAPI, Achilles)
5. trained model enters the export gate: utility, leakage and reproducibility gates plus the canary / LiRA audit
6. leaves the institution only if every gate passes: the model and its audit record (aggregates only, cells of 11 or more)